PRAKTIKUM SISTEM OPERASI

MODUL 4

Membaca Source Code Xinu

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

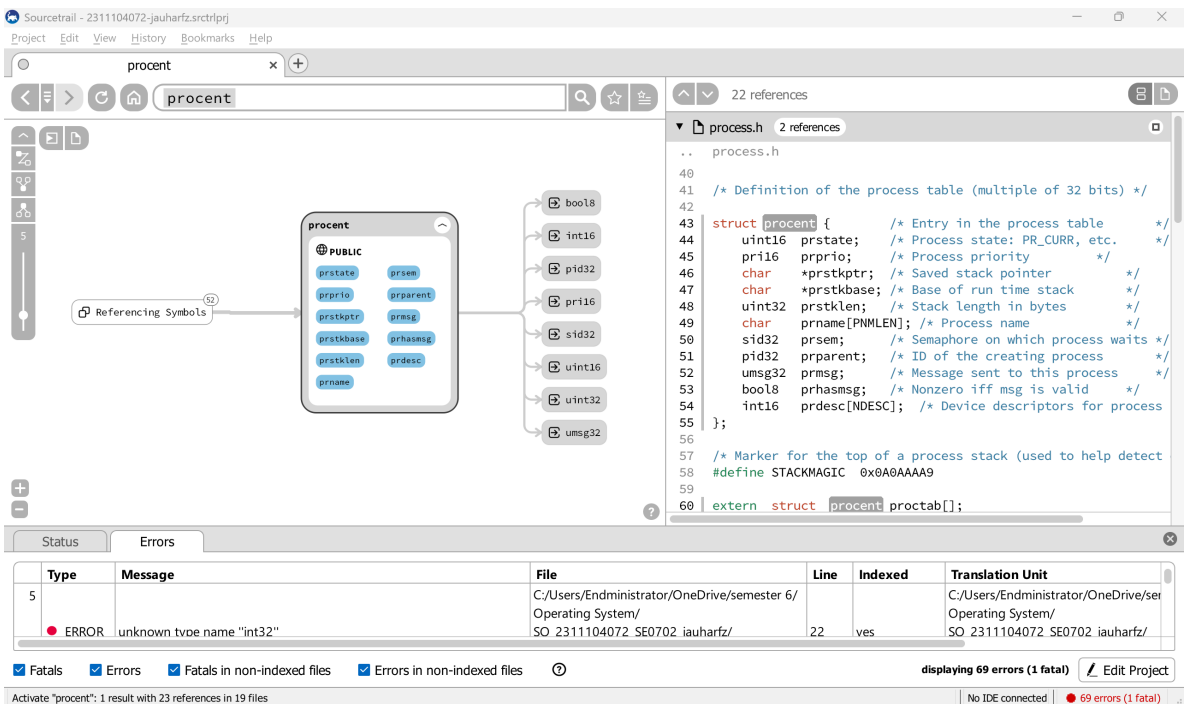
JURNAL

1. Soal 1

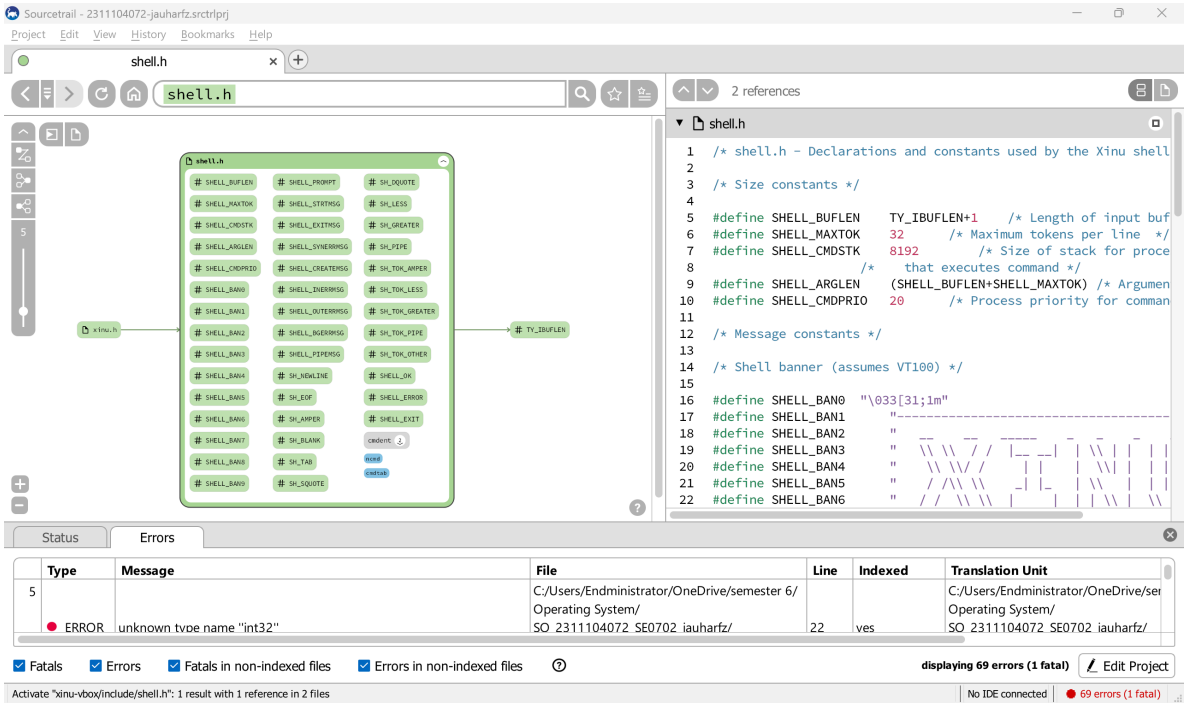
xinu.elf(debug/executable), xinu/compile, 156KB

```
xinu@xinu-develop-end:~$ ls
Desktop  README  Screenshot from 2026-03-13 02-45-57.png  xinu
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ ls
compile  config  COPYRIGHT  DESCRIPTION  device  include  lib  net  shell  system
xinu@xinu-develop-end:~/xinu$ cd compile
xinu@xinu-develop-end:~/xinu/compile$ ls
bin  binaries  ld.script  Makefile  version  xinu.elf  xinu.map
xinu@xinu-develop-end:~/xinu/compile$ ls -lh xinu.elf
-rwxr-xr-x 1 xinu xinu 156K Mar 12 21:42 xinu.elf
xinu@xinu-develop-end:~/xinu/compile$
```

2. Soal 2 & 3



3. Soal 4


Sourcetrail - 2311104072-jauharfz.srctrlprj
Project Edit View History Bookmarks Help
shell.h
shell.h
2 references
shell.h
/* shell.h - Declarations and constants used by the Xinu shell
/* Size constants */
#define SHELL_BUFLEN TY_IBUFLEN+1 /* Length of input buf
#define SHELL_MAXTOK 32 /* Maximum tokens per line */
#define SHELL_CMDSTK 8192 /* Size of stack for proce
/* that executes command */
#define SHELL_ARGLEN (SHELL_BUFLEN+SHELL_MAXTOK) /* Argumen
#define SHELL_CMDPRIO 20 /* Process priority for comman
/* Message constants */
/* Shell banner (assumes VT100) */
#define SHELL_BAN0 "\033[31;1m"
xinu.h
TY_IBUFLEN
Status
Errors
Type
Message
File
Line
Indexed
Translation Unit
5
ERROR
unknown type name "int32"
C:/Users/Endministrator/OneDrive/semester 6/ Operating System/ SO_2311104072_SE0702_jauharfz/
22
yes
C:/Users/Endministrator/OneDrive/ser Operating System/ SO_2311104072_SE0702_jauharfz/
Fatals
Errors
Fatals in non-indexed files
Errors in non-indexed files
displaying 69 errors (1 fatal)
Edit Project
Activate "xinu-vbox/include/shell.h": 1 result with 1 reference in 2 files
No IDE connected
69 errors (1 fatal)

```
  GNU nano 2.2.6              File: include/shell.h                        Modified


/* Shell banner (assumes VT100) */

#define SHELL_BAN0      "\033[31;1m"
#define SHELL_BAN1      "-----------------jauharfz---------------"
#define SHELL_BAN2      "                                         "
#define SHELL_BAN3      "   __   __  _____   _   _   _   _   "
#define SHELL_BAN4      "   \\ \\ / /  |__ __|  | \\ | |  | |  | |   "
#define SHELL_BAN5      "    \\ \\/ /     | |    |  \\| |  | |  | |   "
#define SHELL_BAN6      "    / /\\ \\     _| |_   | \\   |  | |  | |   "
#define SHELL_BAN7      "   / /  \\ \\  |     |  | | \\ |  \\  --  /   "
#define SHELL_BAN8      "   --    --   -----    -   -     ----     "
#define SHELL_BAN9      "-----------------2311104072--------------"
#define SHELL_BAN9      "\033[0m\n"

/* Messages shell displays for user */

#define SHELL_PROMPT    "xsh $ "        /* Command prompt               */
#define SHELL_STRTMSG   "Welcome to Xinu!\n"/* Welcome message          */
#define SHELL_EXITMSG   "Shell closed\n"/* Shell exit message           */
#define SHELL_SYNERRMSG "Syntax error\n"/* Syntax error message         */
#define SHELL_CREATEMSG "Cannot create process\n"/* command error       */
#define SHELL_INERRMSG  "Cannot open file %s for input\n" /* Input err  */
#define SHELL_OUTERRMSG "Cannot open file %s for output\n"/* Output err */

^G Get Help   ^O WriteOut   ^R Read File  ^Y Prev Page  ^K Cut Text   ^C Cur Pos
^X Exit       ^J Justify    ^W Where Is   ^V Next Page  ^U UnCut Text ^T To Spell
```

```
   4447448 bytes of free memory.  Free list:
           [0x0015E320 to 0x0009FFF7]
           [0x00100000 to 0x005FBFFF]
    115497 bytes of Xinu code.
           [0x00100000 to 0x0011C328]
    148488 bytes of data.
           [0x0011FEC0 to 0x001442C7]

Obtained IP address  10.0.3.15   (0x0a00030f)



-----------------jauharfz-----------------
   __   __  _____   _   _   _   _
   \ \  / /  |__ __|  | \ | |  | |  | |
    \ \/ /     | |    |  \| |  | |  | |
    / /\ \     _| |_   | \   |  | |  | |
   / /  \ \  |     |  | | \ |  \  --  /
   --    --   -----    -   -     ----
-----------------2311104072--------------


Welcome to Xinu!


xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```